

BIOL3402 Cell Biology and Cell Technology

Damian Lau

2025-2026 Semester 1

Introduction

This course begins with a discussion of cell biology, covering the Cell, Membrane, Nuclear Transport, Techniques in Plant Cell Culture, Transport across the Plasma Membrane, Ion Channels and Osmotogenetics, Junction, Cytoskeleton, Cellular Adhesion molecules, Cell Migration, Cell Signaling and Techniques in Animal Cell Culture. This course aims to introduce the cellular knowledge. Since this is also a Lab-Lecture based course, this course also provide different culturing knowledge and laboratory skills. Upon completion, students will be familiar with the cell biology and cell technology conceptually and practically.

Contents

Introduction 1

1 Lecture 12 & 13 Cell Coat and The Extracellular Matrix 3

1 Lecture 12 & 13 Cell Coat and The Extracellular Matrix

1. Cell Coat : Cells are surrounded by spaces filled with extracellular matrix
2. Extracellular Matrix (ECM)
 - Proteins + Polysaccharides
 - A major components of connective tissues
 - Different type of matrix and the way they are organized → Different ECM
 - Can become calcified
 - ⇒ Form the rock-hard matrix of **bone & teeth**
 - ⇒ Form the transparent matrix of the cornea
 - Cells secreting ECM
 - (a) Fibroblasts and its derivatives
 - ⇒ Can organize & shape the matrix, allowing migration
 - (b) Chondroblasts in cartilage
 - (c) Osteoblasts in bone
3. Classes of Extracellular Macromolecules that make up ECM
 - (a) Long polysaccharide chains (glycosaminoglycans), which usually covalently linked to protein in the form of proteoglycans
 - (b) Fibrous Protein
 - Structurally : Collagens, Elastin
 - Adhesive : Fibronectin, Laminin
4. The Extracellular Matrix is Made and Oriented by the cells within it
5. Glycosaminoglycan (GAG)
 - Unbranched polysaccharide chain : Composed of repeated disaccharide units

- One of the two sugar in the disaccharide unit : Always a sulfated sugar with amino acid
⇒ N-acetylglucosamine or N-acetylgalactosamine
- High density of negatively charged along the chain
∴ Have sulfated groups and carboxyl groups on the sugar
- Glycosaminoglycan Chains occupy large amounts of space and Form Hydrated Gels

6. Hyaluronic Acid (Hyaluronan)

- Only GAG that does not link with a protein
- Non-sulfated
- Very huge repeated units
- Found in skin, eyes, joints, hair follicles, gums
- Decay + Degrade quickly
∴ No cross linked
- Function 1 : Cell-free space → Cell can migrate
- Function 2 : Lubricant in joint fluid
- Function 3 : Forms a gel in water

7. Proteoglycans

- Protein cores are synthesized in the ER
- Polysaccharides are assembled on the core protein in **golgi**
- Huge size
⇒ Each GAG chain can contain up to 80 sugar residues
- Aggrecan : 20 Amino Acid link to 1 GAG (Can have multiple GAG chain)
⇒ A key component in cartilage
- Decorin : Only 1 GAG chain
- Function 1 : Provide hydrated space around and between cells in ECM
- Function 2 : Bind certain secreted signaling molecules which can affect their function

(a) Enhance the function : FGF $\rightarrow$ Heparin Sulfate

(b) Inhibit the function : TGF- β $\rightarrow$ Decorin

- Function 3 : Serve as **sieves**
 $\Rightarrow$ Regulate the movement of molecules and cells
- Function 4 : Syndecans (Transmembrane Domain) + Integrins act as a receptor
 $\Rightarrow$ For collagen, Fibronectin
- Function 5 : Serglycin is a constitute of intracellular secretory vesicles
 $\Rightarrow$ Help to package and Store secretory molecules

8. Structural and Adhesive fibrous ECM Protein

- (a) Collagens : Provide tensile strength to skin, tendon, connective tissues, blood vessels
- (b) Elastin : Gives tissues their elasticity
- (c) Fibronectin : An adhesive protein that helps cells to attach to the matrix
- (d) Laminin : Major component of basal laminae

9. Collagens

- Provide tensile strength to skin, tendon, connective tissues, blood vessels
- Most abundant protein in mammals
- Collagen fibre : A bundle of fibrils, which in turn is a bundle of collagen molecules
- Each collagen molecule is made of three collagen polypeptide chain

10. Disruption of ECM (No Cross Linking)

- Defect in joints
- Fragile skin

11. Fibrillar (Fibril-forming) Collagens

- Type I, II, III, V, XI collagens
- Fibrillar collagens : Bundles of collagen fibrils
- Fibrils orderly packed into collagen fibre
- Cross Linking : Important for the tensile strength of the fibres $\Rightarrow$ If lack cross linking, collagenous tissues become fragile quickly & lead to degradation

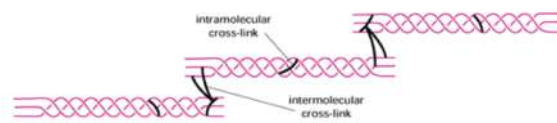


Figure 1: Cross-link

12. Fibril-associated Collagens

- Type IX & XII collagens
- Very small (Smaller than the visible light wavelength)
- Link the fibrils with each other and other components in ECM
- Used to organize the fibrillar collagens

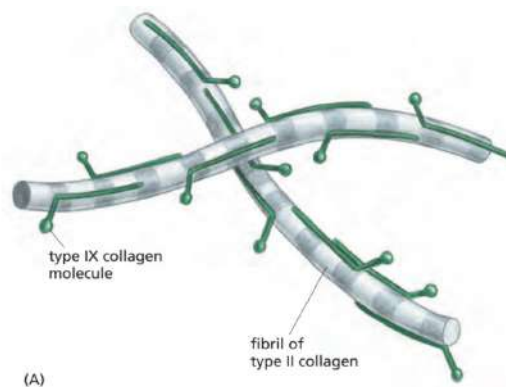


Figure 2: Fibril-associated Collagens

13. Network-forming Collagens

- Type IV collagen

- Constitutes the basal lamina

14. Anchoring Fibrils : Type VII collagen

⇒ Helps the multilayered epithelia anchor to the underlying connective tissue

15. Basal Lamina (Basement Membrane)

- Thin mats of matrix
- Composed of type IV Collagen, Laminin, Perlecan (Proteoglycan) and Entactin
⇒ Anchored to the underlying connective tissue by specialized anchoring fibrils made of type VII collagen
- Under all epithelial cell sheets and tubes
- Surround at muscle cells, schwann cells and fat cells
- Separate two cell sheets and function by **highly selective filter** in kidney glomerulus and lung alveolus

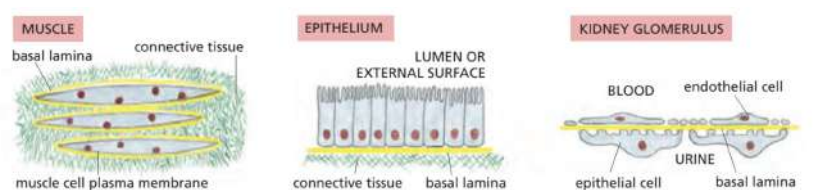


Figure 3: Basal Lamina

- Functions of the Basement Membrane
 - Acts a selective barrier to the movement of cells
 - Separate two cell sheets and function as a highly selective filter in kidney glomerulus and lung alveolus
 - Serve as highway for cell migration
 - Determine cell polarity
 - Induce cell differentiation
 - Organize the protein in adjacent cell membrane
 - Wound healing and tissue regeneration
- The basal lamina is the interface between the epithelium and the connective tissue

⇒ This connection ensures that the entire tissue can withstand mechanical stresses without the layers shearing apart

16. Laminin

- Large flexible complex of three polypeptides (α, β, γ , held together by disulfide bonds)
 - ⇒ Form a shape of asymmetric cross (Cross-like Shape)
 - ⇒ This shape is perfectly suited to its function of linking different elements together
- Multiple Domains and Binding Sites
 - ⇒ Each arm of the cross has specific binding sites (Marked by yellow boxes) for other molecules
- The Three Short Arms
 - ⇒ Typically bind to other components of the basal lamina, such as **other laminin molecules and Nidogen / Entactin** (A protein that helps link laminin to the collagen IV network)
- The Long Arm
 - ⇒ Its tip has binding sites for cell surface receptors
 - ⇒ Most importantly integrins and dystroglycan on the basal surface of epithelial cells

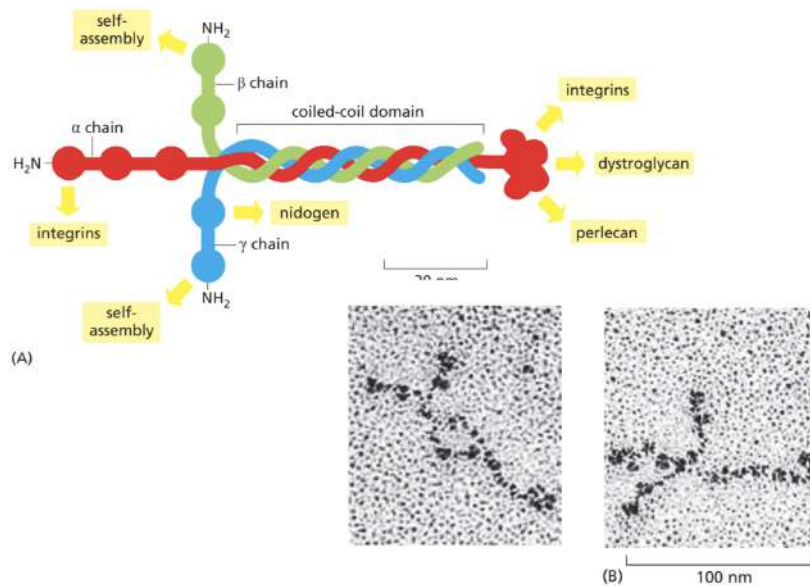


Figure 4: Laminin

- Can contain many functional domains : Bind to One Type IV collagen + One Heparan Sulfate + One Entactin + > 2 Integrins

17. Entactin

- Act as an additional link between laminin and Type IV collagen
⇒ Sulfated, Highly Negatively Charged, Carboxyl group have negative charge
- Essential for connecting the two primary structural networks together
- Entactin (Nidogen) is a sulfated glycoprotein that acts as a key bridge or linker in the basal lamina
- Laminin : The top layer that interfaces with the cells
- Type IV Collagen : The bottom layer that provides strength
- Entactin (Nidogen) : The adhesive layer in the middle

18. Elastic Fibre

- Protein networks that allow tissues to stretch and then return to their original shape without damage
- Composed by elastin protein
- Function : This gives tissues like blood vessels and lungs the ability to stretch and recoil endlessly throughout a lifetime

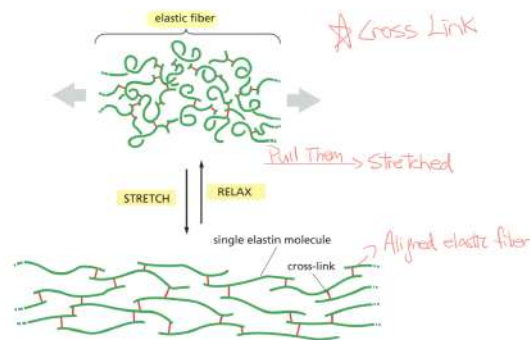


Figure 5: Elastic Fibre

- At least 5 times more extensible than a rubber band
- Collagen fibers mix with the elastin fibers limit the extent of stretching and prevent the tissue from tearing
- Highly hydrophobic protein that is rich in proline and glycine
- Crosslinked at lysine
⇒ Creating a permanent, stable connection between the four elastin chains
- Elastin fiber is made of elastin
- Microfibrils that are composed of several glycoproteins
- Application : Cosmetic Treatments

19. Fibronectin

- Contain many function domain : Binding to collagen, heparin, and integrins
- Form disulfated bond at the C-terminal → Can form a fibronectin dimer

- Synergy Sequences : Important for binding to integrins on cell surfaces

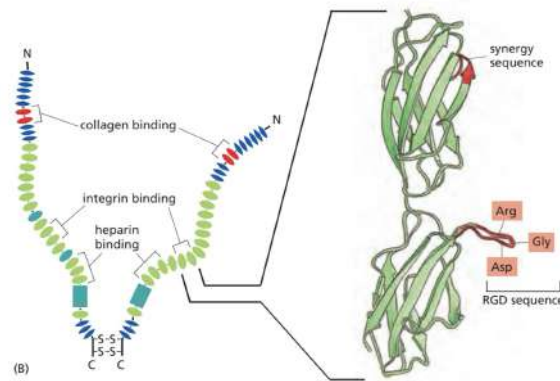


Figure 6: Fibronectin

- Fibronectin is a versatile adhesion protein in the ECM
- Encoded by a single large gene (50 Exons)
- Alternative RNA splicing produce 20 mRNA species in human multiple
 - ⇒ Translated into multiple forms of fibronectin
 - ⇒ Produce the most suitable form of fibronectin required by the tissue
- Plasma Fibronectin : Highly soluble and circulates in the blood
 - ⇒ It participates in wound healing, blood clotting and phagocytosis
- Fibronectin Filaments : Insoluble fibronectin deposited on cell surface
 - ⇒ It brings to cell migration

20. Cell Migration (Importance)

- Immune Response
 - ⇒ White blood cells must migrate from the bloodstream to sites of infection or injury to destroy pathogens (Macrophage chasing down a bacteria)

- Tumor Metastasis : Bad Migration
- Development
 - ⇒ Cells must travel long distances to reach their correct positions and form tissues and organ during embryonic development
- Wound Healing (A slow process & not random)
 - ⇒ Have Blood Clotting
 - ⇒ Platelets Aggregates

21. Immune Cell : Eat Bacteria

- (a) Remodel matrix and secrete more matrix
- ⇒ Allow more cell can come in
 - ⇒ Then, start to heal up the mound

22. Cell Migration (Directional Persistence)

- High / Low (**Constant Switching Direction**)
- Directed
 - ⇒ E.g. Keretocyte = Very Directed Migration (Very Persistent Migration)
- Random

23. Cell Migration (Dimensionality)

- 1D (Line) : Can only move along the line (very aligned)
- 2D (Plane) : Move Front, Back, Left, Right
 - ⇒ E.g. Basement Lamina (Laminin)
- 3D (Matrix) : Cell is embedded + Stick the matrix to collagen

24. Cell Migration (Migration Mode)

Feature	Amoeboid Migration	Mesenchymal Migration	Collective Migration
Cell Shape	Rounded, blebbing	Elongated, spindle-shaped	Cohesive sheet, strand, or cluster
Adhesion	Weak to ECM	Strong to ECM (focal adhesions)	Strong cell-cell junctions
Speed	Fast	Slow	Variable (slow to moderate)
ECM Degradation	No (Squeezes through)	Yes (Clears a path)	Yes (by leader cells)
Key Example	White Blood Cells	Fibroblasts in Wound Healing	Epithelial Sheet Repair

Figure 7: Migration Mode

25. Amoeboid vs Mesenchymal Migration

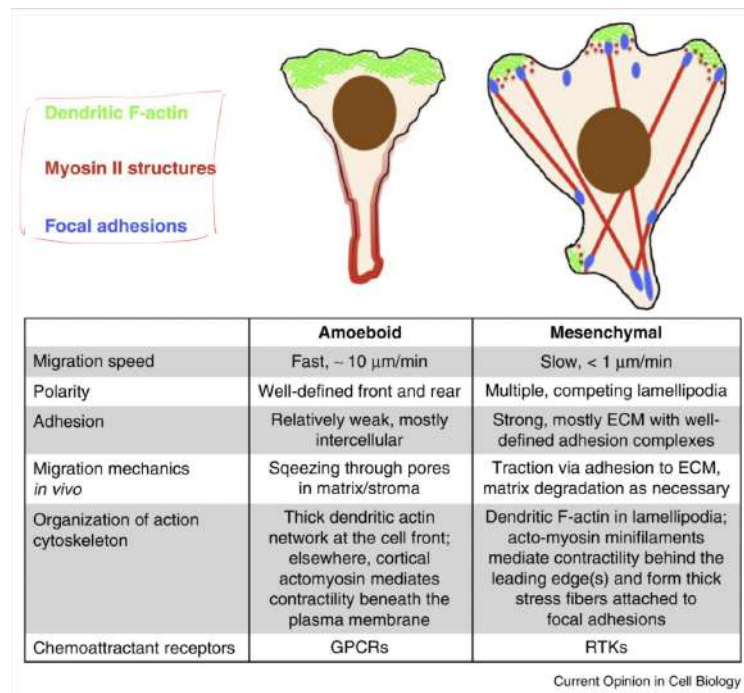


Figure 8: Amoeboid vs Mesenchymal Migration

26. Migration : Directed by cues (Random)

- Can move very fast (Have new biological migration guided by migration)
- Crucial to some biological functions
 - ⇒ Involve the degradation of ECM macromolecules
- Cell secretes proteases (serine proteases and metalloproteases)
 - Breakdown ECM macromolecules locally
 - ∴ Cell migration can be enabled by a relatively small amount of proteolysis

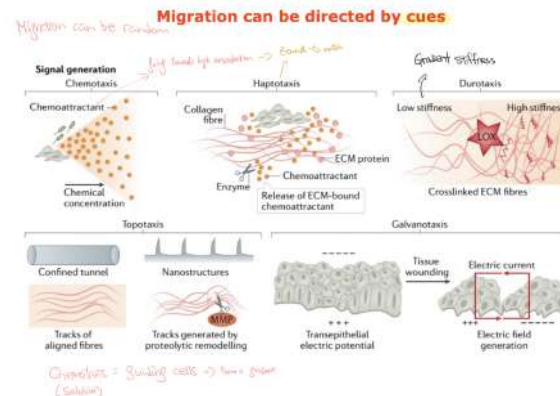


Figure 9: Migration

27. Proteolysis : Urokinase Receptor

- ⇒ For matrix remodeling & usually tethered
- ⇒ Usually display on the surface

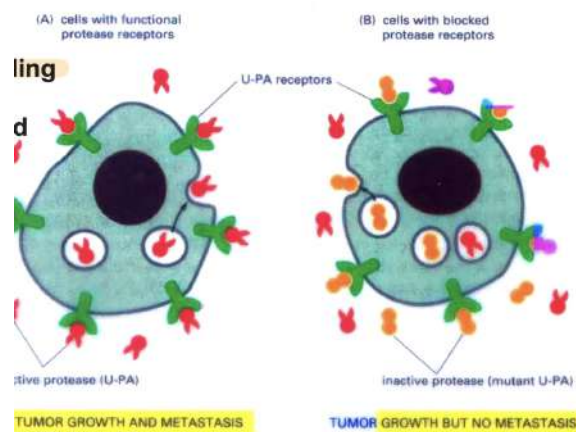


Figure 10: Proteolysis

28. Actin organization within the cell

- Very dynamic
- organized differently in cells
- Type of Bundle
 - (a) Contractile Bundle : Stress Fibre
 - ⇒ Contractile Force : Driving Force of cell migration

- (b) Gel-like Network : Cell Cortex
- (c) Dendritic Network : Lamellipodium
- (d) Tight parallel Bundle : Filopodium

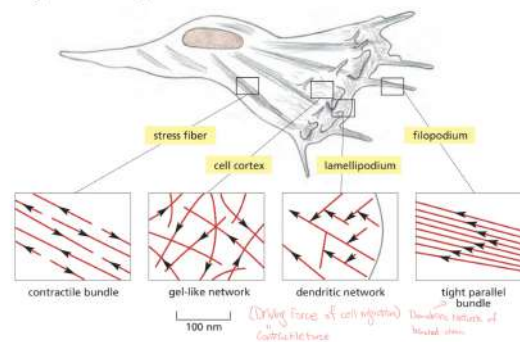


Figure 11: Bundle

- Actin Structures

- (a) Lamellipodia : Thin sheet-like protrusions that contain a branched network of actin
- (b) Filopodia : Thin finger-like protrusions that are composed of tight parallel bundles of actin
- (c) Stress Fibres : Anti-parallel actin filaments associated with myosin II bundles
- (d) Arp 2/3 : Mediates the formation of branched actin
 - ⇒ Start to grow a new filaments
 - ⇒ Critical for the formation of branched actin
- (e) Method to grow more
 - Remove the capping
 - Have an ARP2/3 → Branch Development

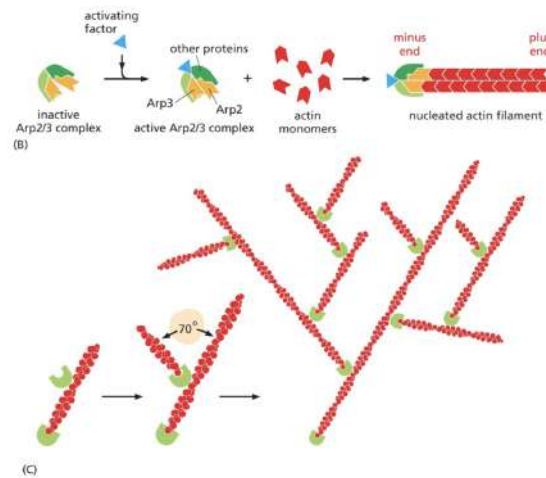


Figure 12: Arp 2 / 3

- (f) *Listeria monocytogenes* : Hijack Arp2/3 actin polymerization machinery
- (g) Fascin : bundles actin filaments together
- (h) Capping Proteins : Cap growing actin filaments preventing elongation

29. Basic steps in Cell Migration

- (a) Protrusion of the leading edge
- (b) Adhesion to the ECM
- (c) Cell body translocation
- (d) Retraction of the trailing edge

30. Pathway : Cell Migration → Active Polymerizing Actin + Extending → help attach to cell matrix

31. Filopodia Initiation

- mDia / ENA / VASP selectively elongate branched actin filaments
- I-BAR domain proteins promote membrane deformation
- Fascin crosslinks and stabilize actin filaments

32. Integrins & Focal adhesions in Migration

(a) Integrins

- Cell Surface Receptors (α / β)
 $\Rightarrow$ Bind specifically to the ECM
- Form linkage between cytoskeleton and ECM

(b) Focal Adhesions

- Several integrins can group together to form nascent adhesions
- Can mature to form more stable adhesions

33. Actin network is rapidly remodeled to allow cells to migrate :
 Remove the cap

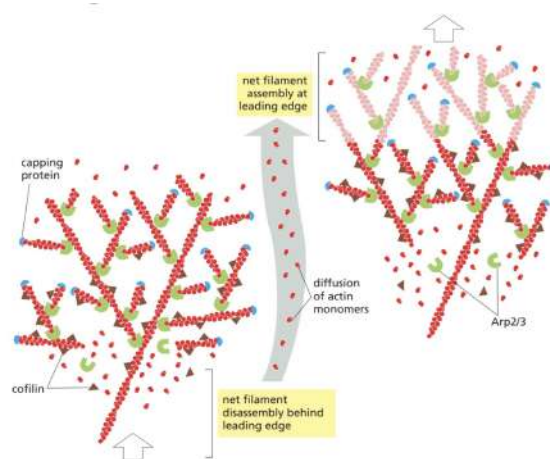


Figure 13: Net Filament

34. Myosin : Force generating molecular motors of the cell
 $\Rightarrow$ Coil-Coil Helices $\rightarrow$ Binding to either actin / microtubules

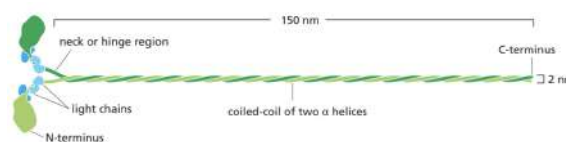


Figure 14: Myosin II

35. Myosin-Actin Cross-Bridge Cycle

(a) Attached

- i. State : The myosin head is bound tightly to actin without nucleotide bound & it is a **rigid state**
- ii. In the Cell : This state is responsible for rigor mortis after death, when ATP is depleted and the myosin heads cannot detach from actin
- iii. It is a very short life, terminated by the binding of a molecule of ATP

(b) Released (Detachment)

- i. Trigger : A molecule of ATP binds to a pocket on the back of the myosin head
- ii. Action : The binding of ATP causes a small conformational change in the myosin head → **Reduces its affinity for actin**
- iii. This forces the myosin head to detach from the actin filament
- iv. In reality, the head probably remains very close to the actin

(c) Cocked (Priming / Recovery Stroke)

- i. Trigger : The myosin head **hydrolyzes the ATP** into **ADP and inorganic phosphate (P_i)**
- ii. The products (ADP and P_i) remain tightly bound to the protein head
- iii. Action : The energy from hydrolysis is stored in the protein, causing a large conformational change
⇒ Cocks the myosin head into a high-energy, primed position
- iv. The neck (lever arm) swings forward, readying the head to bind further along the actin filament

(d) Force-Generating

- i. Trigger : The cocked myosin head now binds weakly to a new site on the actin filament.
- ii. This binding triggers the **release of the inorganic phosphate (P_i)** from the head

- iii. Action : The release of P_i causes the myosin head to undergo a dramatic shape change
⇒ The **POWER STROKE** : The central force-generating step
 - iv. The neck (lever arm) snaps back to its original, bent conformation, pulling the actin filament along with it
 - v. This is the step where force is generated
 - vi. Completion : At the end of the power stroke, the **ADP is released** + Myosin head is once again tightly bound to actin in the **rigor state (Step 1)**, ready to begin a new cycle.
- (e) Attached (Back to Step 1)
- i. Myosin head is again locked tightly to the actin filament in a rigor configuration
 - ii. The head has moved to a new position on the actin filament
- (f) Myosin Head : Only actually bound with actin 5% of cycle time
- The myosin head spends most of its cycle detached from actin (In the "Cocked" state)
 - Function : Allows the filament to slide smoothly without all heads being stuck at once
- (g) The Direction the Myosin Motor Moves & Filament Movement
- Normal Case : The myosin head move toward the **Plus End** (Barbed End) of the actin filament
⇒ An actin filament is a one-way street
 - Muscle Contracting : The actin filaments are pulled toward the **Minus End** (Pointed end)
 - Arrangement 1 : Actin filaments' Plus Ends pointing away from the center
 - Arrangement 2 : Their Minus Ends pointing at the center

- When the myosin heads on both sides walk toward the Plus End
 - ⇒ They pull the actin filaments toward the center
 - ⇒ This means the actin filaments are being pulled toward their own Minus Ends

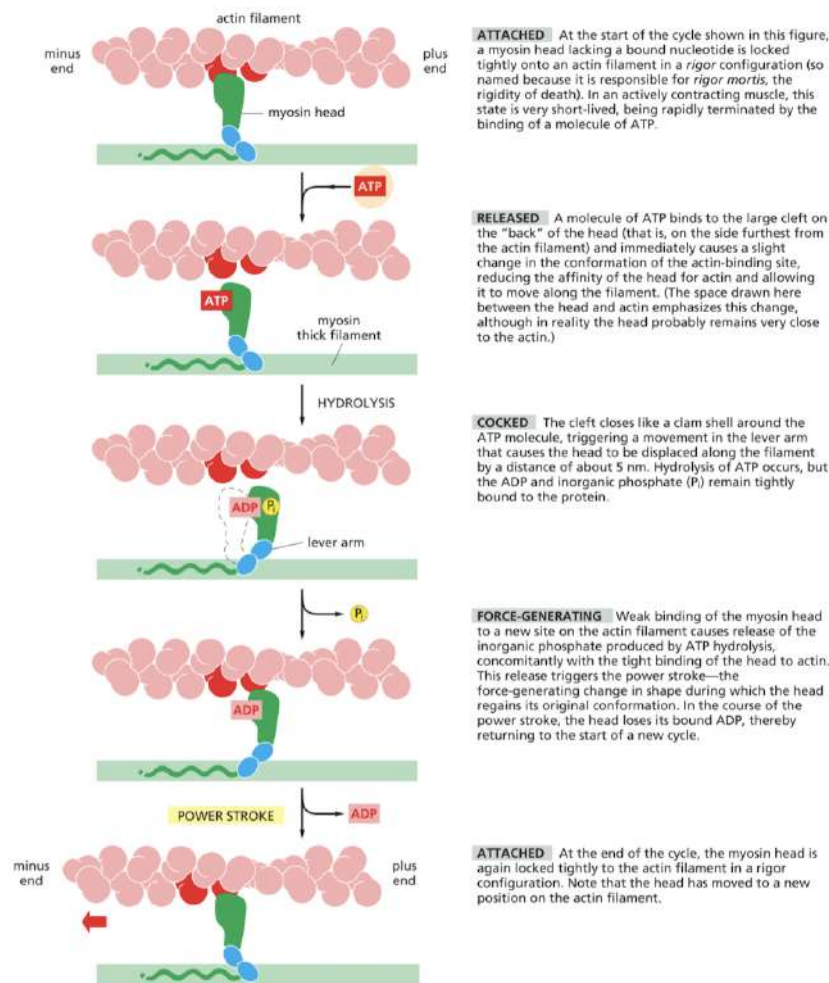


Figure 15: Muscle Contraction

- (h) The contracted force is generated in the power stroke, occurs when the myosin head re-bounded to the actin filament, bring the actin moving toward minus end
- ⇒ Caution : Not the Bounding Reaction cause the contraction and the movement, but the power stroke

36. Actin Staining

- Key regulators of the actin cytoskeleton : Small Rho family GTPases
- Example 1 : Quiescent Cells
- Example 2 : Cdc42 Activation
- Example 3 : Rac Activation
- Example 4 : Rho Activation
- All example actin straining are specific to each others
⇒ But have same mechanism